

Reviewer Pack — Minimal Rank of Tropicalized Affine Sheaves vs Communication...

Entry 3d9cc1fcfc83 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-23 16:37:49 UTC

Minimal Rank of Tropicalized Affine Sheaves vs Communication Complexity for Disjointness

Entry ID: 3d9cc1fcfc83 Verdict: INCONCLUSIVE Recorded: 2026-05-23 16:37:49 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry (Affine Sheaves) **Field B** (complexity object): Communication Complexity (Disjointness)

Statement:

[‘The minimal rank of the tropicalization of an affine sheaf associated with a disjointness instance, $\tau(T)$, is lower bounded by the randomized communication complexity of the disjointness instance, $C(\text{DISJ}_n)$, such that $\tau(\text{DISJ}_n) = \Omega(n)$.’, ‘For all disjointness instances of size $n \leq 40$, there exists an affine sheaf whose tropicalization has a minimal rank of at least cn (for some constant $c > 0$ and random seeds), where $C(\text{DISJ}_n)$ is the randomized communication complexity for the instance.’, ‘A counterexample to this conjecture would be an affine sheaf with a tropicalization of minimal rank less than cn for all instances $n \leq 40$, where $C(\text{DISJ}_n) = o(n)$.’]

Rationale (proposer’s reasoning):

[‘Tropical geometry provides a powerful framework for studying the complexity of computational problems by encoding geometric structures into algebraic objects. Affine sheaves, which generalize the concept of divisors on curves, offer a rich source of invariants that can capture the essence of computational problems. By linking the minimal rank of tropicalized affine sheaves to communication complexity, this conjecture aims to uncover a deeper connection between geometry and computation.’, ‘The connection with disjointness is motivated by its fundamental nature in communication complexity theory. If the conjecture holds, it would suggest that geometric invariants derived from the structure of affine sheaves can be leveraged to provide lower bounds on communication complexity, which has been an active area of research for several decades.’]

Taxonomy category: COMM_DISJ (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 4c5b70983fb6296d

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all $n \leq 40$ and across 30 random seeds, the ratio of the estimated randomized communication complexity to the minimal rank of the tropicalized affine sheaf is greater than or equal to a constant c .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 8 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - tropical geometry AND affine sheaves AND communication complexity AND disjointness - minimal rank tropicalization affine sheaves OR communication complexity disjointness - disjointness instances communication complexity lower bound minimal rank tropicalization

Top relevant hits considered: - [<http://arxiv.org/abs/1207.2443v2>] Tropical Teichmuller and Siegel spaces - [<http://arxiv.org/abs/2303.05618v4>] Tropical Geometry, Quantum Affine Algebras, and Scattering Amplitudes - [<http://arxiv.org/abs/0802.4323v1>] Non-singular affine surfaces with self-maps - [<http://arxiv.org/abs/1403.8106v1>] Recent advances on the log-rank conjecture in communication complexity - [<http://arxiv.org/abs/1102.2932v2>] Applications of Monotone Rank to Complexity Theory - [<http://arxiv.org/abs/2401.14623v1>] Structure in Communication Complexity and Constant-Cost Complexity Classes - [<http://arxiv.org/abs/1806.02734v3>] Spectral lower bounds for the orthogonal and projective ranks of a graph - [<http://arxiv.org/abs/cs/0111062v2>] One-way communication complexity and the Neciporuk lower bound on formula size

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_disjointness_instance(n):
        return [random.choice([0, 1]) for _ in range(n)]

    def communication_complexity(instance):
        n = len(instance)
        cc = 0
        for i in range(n):
```

```

        if instance[i] == 1:
            cc += 1
    return cc

def tropical_rank(instance):
    n = len(instance)
    rank = 0
    for i in range(n):
        if instance[i] == 1:
            rank += 1
    return rank

n = random.randint(5, 40)
instance = generate_disjointness_instance(n)
cc = communication_complexity(instance)
mrts = tropical_rank(instance)

ratio = cc / mrts if mrts != 0 else float('inf')

return {
    "metric_name": "Ratio of Communication Complexity to Minimal Rank",
    "metric_value": ratio,
    "instances_tested": 1,
    "conjecture_holds": ratio >= n * 0.5, # Example threshold for c
    "counterexample": f"cc={cc}, mrts={mrts}" if not conjecture_holds else ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"cc={results[0]['metric_value']}, mrts={results[0]['mrts']}\"")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5364bfa0.py", line 61, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5364bfa0.py", line 52, in run_trial
    "counterexample": f"cc={cc}, mrts={mrts}" if not conjecture_holds else ""
                                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
NameError: name 'conjecture_holds' is not defined
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from verifying the conjecture's support or falsification. | next: Investigate and fix the crash in the test code to proceed with the verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	16582
2	preregistration	ollama_remote	glm4:latest	0	0	9165
3	novelty	ollama_remote	glm4:latest	0	0	8025
4	novelty	ollama_remote	glm4:latest	0	0	9429
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11081
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14454
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13016
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7475
9	judge	ollama_remote	glm4:latest	0	0	13313

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 102538 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in *research/audit/3d9cc1fcfc83.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/3d9cc1fcfc83.jsonl* for verification of provenance

4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/3d9cc1fcfc83.tar.gz (if generated)